CUformer: A Channel-wise Unet-based Transformer for Semantic Segmentation of Remote Sensing

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***Abstract*—Remote sensing semantic segmentation is applied to land cover mapping and urban change detection. Recently, some methods introduced CNN and Vit hybrid methods to solve the remote sensing semantic segmentation. Furthermore, to reduce the computation cost, they mainly introduce the window-self-attention or not the villain self-attention to extract the global semantic feature. However, the window-self-attention ignores the channel interaction information. To fix the problem, we propose Channel-wise Unet-based Transformer (CUformer). The CUformer introduces channel attention to add the channel interaction information into the window-self-attention. Furthermore, previous works have proved that semantic segmentation needs to fine-grain the global semantic feature. Therefore, we design a simple fine-grained head, which introduces a local feature to fine grain the global semantic feature. Extensive experiments show that our model achieves state-of-the-art performance in Potsdam and Vaihingen datasets.**

Keywords-remote sensing; semantic segmentation; channel; Transformer; CNN

I. INTRODUCTION

Semantic segmentation is a pixel-wise task classification task that assigns every pixel a class. Their application in remote sensing images is one of the current research focuses. Semantic segmentation of remote sensing images is of great importance in earth observation, land use, land cover, and mapping applications [1]. Today, with the rapid development of remote sensing acquisition technology, remote sensing images have the characteristics of high resolution. High-resolution remote sensing images contain a large amount of surface object information, showing diversity and complexity, which is difficult to the global semantic feature with a good trade-off between accuracy and computation cost. In order to fix the problem, A common method is to introduce the window-self-attention [2] to extract the global semantic feature while having a low computation cost. But the window self-attention ignores the channel interaction information in the window self-attention.

To fix the problem, we propose Channel-wise Unet-based Transformer (CUformer) to address the above problems while a good trade-off. Furthermore, previous works have shown that global semantic feature needs to be fine-grained [3] for remote sensing semantic segmentation. Therefore, we design a simple fine-grained head.

The CUformer is an Unet-based architecture [4], which consists of a CNN encoder and a proposed Transformer-based decoder, the CNN encoder can extract the local feature. The transformer-based decoder introduces channel window-self-attention to add channel interaction information in the global semantic feature. And our fine-grained head introduces the local feature to the fine-grain global semantic feature. Besides all, we introduce the auxiliary loss [5] to add the performance with a tiny extra train computation cost.

We conduct the experiment on the Potsdam and Vaihingen datasets to show that our method can achieve a state-of-the-art performance with a good trade-off.

In summary, our contribution can be listed as:

- We propose a channel window self-attention to add the channel interaction information into the global semantic feature while a good trade-off.
- We propose an SFHead to fine-grain the global semantic feature.
- Our method can achieve state-of-the-art performance.

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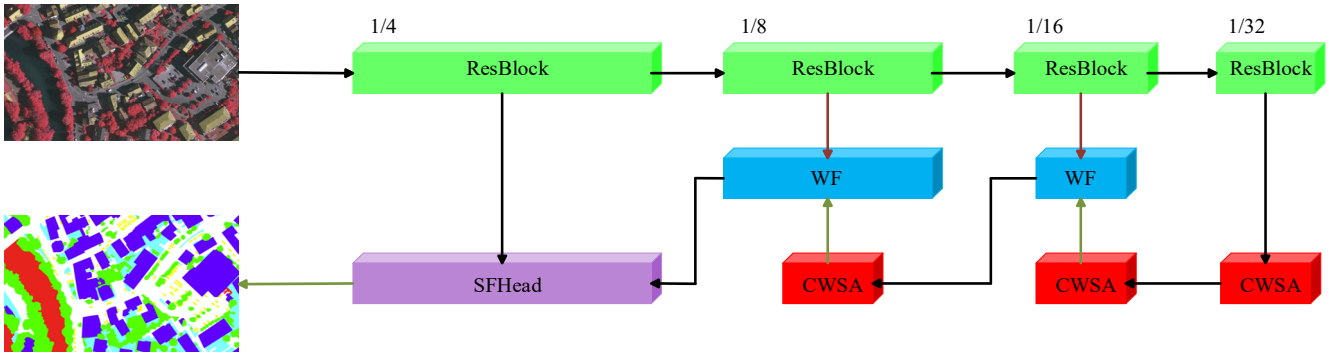


Fig. 1 An overview of the CUformer

II. METHOD

This section details the structure of CUformer. Firstly, the overall structure of the model is introduced, and then each module of the model is introduced in detail.

A. Overview of the Proposed CUformer

As shown in **Fig. 1**, the CUformer consists of a CNN-based encoder and a Transformer-based decoder. In order to reduce the inference time, we choose the pre-trained ResNet18 as our backbone, due to the ResNet18 has proven its effectiveness and good efficiency in dealing with a wide range of real-time semantic segmentation tasks. In CUformer, the encoder is composed of four CNN blocks to extract the local features of different scales.

We feed image $I \in \mathbb{R}^{H\times W\times 3}$ into the backbone to get the local $I_1 \in \mathbb{R}^{\frac{H}{4}\times\frac{W}{4}\times 3}$, $I_2 \in \mathbb{R}^{\frac{H}{8}\times\frac{W}{8}\times 3}$, $I_3 \in \mathbb{R}^{\frac{H}{16}\times\frac{W}{16}\times 3}$, $I_4 \in \mathbb{R}^{\frac{H}{32}\times\frac{W}{32}\times 3}$ feature. Thanks to the high efficiency of ResNet18, the training and inference time will be reduced largely. Subject to the limited receptive field of convolution, it cannot obtain global context information. In order to extract the global semantic feature, we propose a Transformer-based decoder. We introduce a convolution layer on the output of backbone I_4 to get the input of decoder I_d. Then we introduce a channel window self-attention (CWSA) block to get the global semantic feature with channel interaction information G_i. Then we introduce a weight-fused (WF) module to fuse the local feature and global semantic feature to get the local-global context F_i. And we repeat the above process three times in total. Finally, we introduce a simple fine-grained head (SFHead) to fine grain feature.

B. Channel Window Self-attention module

Transformer [6] can obtain global semantic information well. And window self-attention can reduce the computation cost largely. However, for the traditional window-self-attention, it ignores the channel interaction information in the global semantic information. Therefore, in order to reduce the computational complexity, we propose a channel window self-attention (CWSA) module. As shown in **Fig. 2**. CWSA combined window self-attention and channel attention to add channel interaction information into the global semantic feature. For the window self-attention, we get Q, K, and V by convolution, which stands for query, key, and value, respectively. Then, we add channel attention on the V to add channel interaction information. Formally, the calculation process can be expressed as:

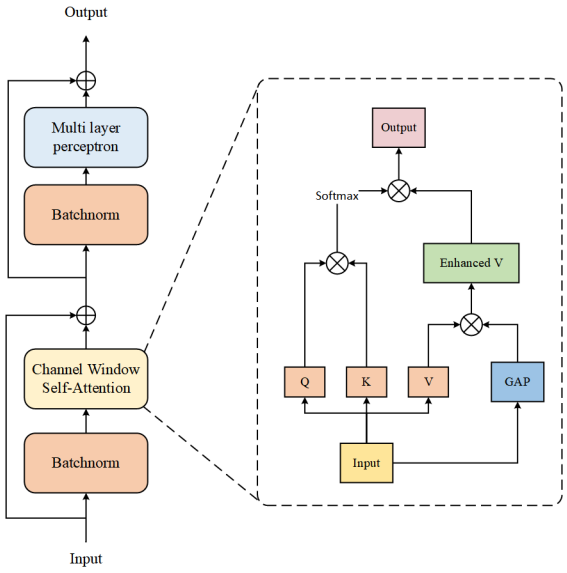


Fig. 2 Illustration of the channel window self-attention (CWSA)

$$Attention(Q,K,V^{'}) = SoftMax(\frac{QK^{T}}{\sqrt{d}} + B)V^{'} \quad (1)$$

$$V^{'} = V \cdot Ca \quad (2)$$

$$Ca = Act(GAP(ConvBN(Input))) \quad (3)$$

where $V^{'}$ represents the enhanced V. It has channel integration information. Softmax means softmax activation function. ConvBN represents 1×1 convolution with batch norm layer. GAP represents the global average pooling operation. Act means sigmoid activation function.

In this way, we can extract the global semantic feature G_i with channel interaction information.

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C. Weight Fused module

In order to filter and fuse the local feature and global semantic feature. We propose the WF module. As shown in

Fig. 3, the WF module introduces 2 learnable parameters to get the weight, which can filter the local feature L_i and global feature G_i. Then we add them to get the global-local context information. Formally, the process of the WF module can be written as:

$$O_{wf} = fw_1 \cdot Conv(L_i) + fw_2 \cdot ups(G_i) \tag{4}$$

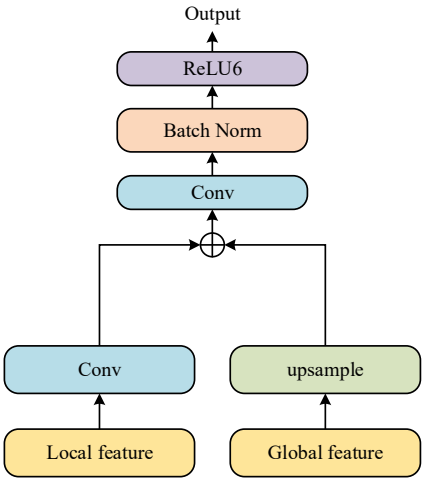


Fig. 3 Illustration of the weight fused (WF)

Where O_{wf} represents the output of WF. $fw1$, $fw2$ represent 2 learnable parameters. ups represents upsampling.

D. Simple Fine-grained head module

Due to upsample the local-global context information, detailed information will be lost in the process of upsampling. Therefore, we need to fine-grain the local-global feature. The feature map of CWSA after three times makes up for this shortcoming, so we directly add the two to improve the segmentation speed. However, the simple addition operation is fast, but it will lose accuracy, so we propose SFHead to solve the problem caused by the semantic gap.

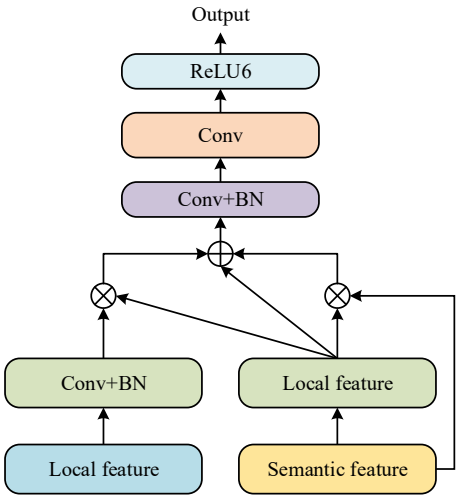


Fig. 4 Illustration of the simple fine-grained head (SFHead)

As shown in **Fig. 4,** we use two channels to enhance the semantic information of the spatial dimension and the channel dimension. We first upsample the fused feature map 2 times and then enhance its spatial and channel semantic information through two branches.

The process can be expressed as：

$$O_{sf} = p \cdot O_{wf} + c \cdot O_{wf} \tag{5}$$

where p represents the spatial coefficient, which is formed by the input passing through a 3×3 convolution and a sigmoid function, and c represents the channel coefficient, which is formed by the input passing through average pooling and 1×1 convolution, and then passing through ReLU6 and sigmoid function to get the output of SFHead.

After the fusion feature map is enhanced by the information of the two paths, we directly add the results of the two paths, and after a deep separable convolution and BN, the results are obtained by residual connection.

III. EXPERIMENT

A. Datasets

The remote sensing scene datasets we used in this experiment are Vaihingen and Potsdam.

The Vaihingen dataset contains 33 remote sensing images of various sizes. These images are very classic. Each image is extracted from a larger top-level ortho image. The image selection process avoids data loss. Spatial resolution of top image and DSM is 9 cm. The remote sensing image format is an 8-bit TIFF file, which consists of three bands of near-infrared, red, and green. DSMs are single-band TIFF files with gray levels (equivalent to DSM heights) encoded as 32-bit floating point values. The annotation data is grouped into six categories: impervious surfaces, buildings, low vegetation, trees, vehicles, and clutter/background. Only TOP image patches without DSM and NDSM are used in the experiments. Use IDs: 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38 to test and train

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the remaining 16 pictures for Image Blocks are cut into blocks of 1024 x 1024 pixels.

TABLE 1 Ablation study of each component of the CUformer

Dataset	Method	mIoU
Vaihingen	Baseline	77.1
	Baseline+CWSA	82.3
	Baseline+CWSA+WF	83.3
	Baseline+CWSA+WF+SFHead	83.6

The Potsdam dataset contains 28 images of the same size as the above image, and the DSM spatial resolution is 5cm. Similar to Vaihingen, the dataset consists of 3-band remote sensing TIFF files and 1-band DSM files. Each remote sensing image covers the same area.

Therefore, remote sensing imagery and DSM are defined on the same reference system (UTM WGS84). Each image has an affine transform file that transforms the image into smaller images as needed.The number of feature classes in the labeled data corresponds to the number of classes in the Vaihingen dataset. ID: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, 7_13 are used for testing, and the remaining 23 pictures (excluding the wrongly titled picture 7_10) are used for training. In the experiments, the original image patch is divided into blocks of 1024x1024 pixels.

B. Experimental settings

All models in our experiments are implemented on two NVIDIA Tesla P40 GPUs using the PyTorch framework. The base learning rate is fixed at 6e-4. The learning rate is adjusted using a cosine policy.

For the Vaihingen and Potsdam datasets, images are randomly divided into segments of size 512 × 512. Training used refinement techniques such as random scaling ([0.5, 0.75, 1.0, 1.25, 1, 5]), random vertical rotation, random horizontal rotation.rotation and rotation. The training epoch is set to 100 and the stack size is set to 16. The testing phase uses random and multi-step boosting.

The metrics used in our experiments fall into two main categories. The first is to evaluate network accuracy, such as overall accuracy (OA), average F1 score (F1), and average interconnectivity (mIoU). This is followed by floating-point numbers (flops) to evaluate complexity, frames per second (FPS) to evaluate speed, memory consumption (MB) to evaluate memory consumption, and model parameters (M) to evaluate network scale.

TABLE 2 Results on the Vaihingen test set are quantified against state-of-the-art lightweight networks. The best value in the column is shown in bold.

Method	Backbone	Imp.surf.	Building	Lowveg.	Tree	Car	MeanF1	OA	mIoU
SwiftNet[7]	ResNet18	92.2	94.8	84.1	89.3	81.2	88.3	90.2	79.6
ABCNet[8]	ResNet18	92.7	95.2	84.5	89.7	85.3	89.5	90.7	81.3
BoTNet[9]	ResNet18	89.9	92.1	81.8	88.7	71.3	84.8	88.0	74.3
BANet[10]	ResT-Lite	92.2	95.2	83.8	89.9	86.8	89.6	90.5	81.4
EaNet[11]	ResNet18	91.7	94.5	83.1	89.2	80.0	87.7	89.7	78.7
ShelfNet[12]	ResNet18	91.8	94.6	83.8	89.3	77.9	87.5	89.8	78.3
Segmenter[13]	Vit-Tiny	89.8	93.0	81.2	88.9	67.6	84.1	88.1	73.6
SPANet[14]	ResNet50	92.1	94.9	82.8	89.1	88.1	89.4	90.0	81.1
DCFAM[15]	Swin-S	**93,6**	**96.2**	**85.8**	90.4	87.6	90.7	**91.6**	83.2
CUformer	ResNet18	93.1	95.7	85.1	**90.8**	**90.0**	**91.0**	91.3	**83.6**

TABLE 3 Results on the Potsdam test set are quantified against state-of-the-art lightweight networks. The best value in the column is shown in bold.

Method	Backbone	Imp.surf.	Building	Lowveg.	Tree	Car	MeanF1	OA	mIoU
SwiftNet[7]	ResNet18	91.8	95.9	85.7	86.8	94.5	91.0	89.3	83.8
FANet[16]	ResNet18	92.0	96.1	86.0	87.8	94.5	91.3	89.8	84.2
Segmenter[13]	Vit-Tiny	91.5	95.3	85.4	85.0	88.5	89.2	88.7	80.7
BANet[10]	ResT-Lite	93.3	96.7	87.4	89.1	96.0	92.5	91.0	86.3
EaNet[11]	ResNet18	92.0	95.7	84.3	85.7	95.1	90.6	88.7	83.4
ShelfNet [12]	ResNet18	92.5	95.8	86.6	87.1	94.6	91.3	89.9	84.4
SwinUperNet[17]	Swin-Tiny	93.2	96.4	87.6	88.6	95.4	92.2	90.9	85.8
SPANet[14]	ResNet50	91.6	94.9	87.1	88.9	94.1	91.3	89.7	84.2
WiCoNet[18]	ResNet50	92.5	96.5	87.0	87.3	95.1	91.7	90.2	84.9
CUformer	ResNet18	**93.8**	**97.1**	**87.8**	**89.4**	**96.6**	**92.9**	**91.4**	**86.9**

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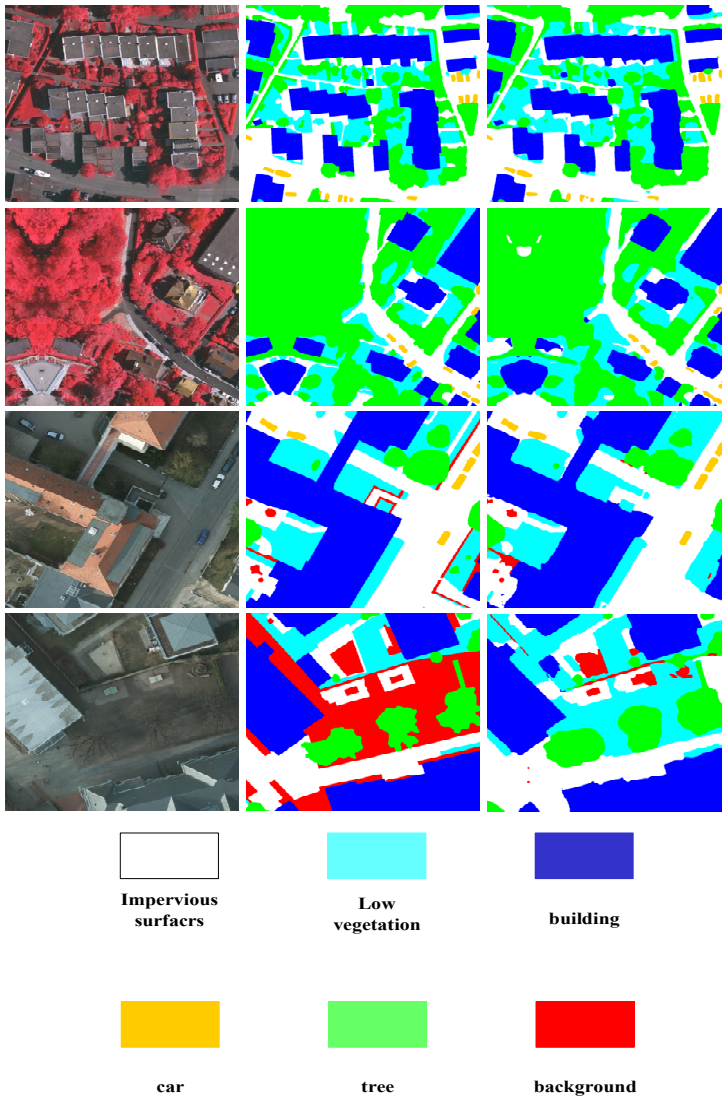


Fig. 5 Visualization results of the Vaihingen and Potsdam test set. The first column represents the input RGB image, the second column represents the absolute ground truth, and the third column shows the proposed results of CUformer segmentation.

C. *Ablation study*

In order to verify the effectiveness of each module of our model, we have done a series of ablation studies on the Vaihingen dataset. As shown in **TABLE 1**, after adding CWSA to the decoder, the performance of the network has been greatly improved. Compared with the baseline, mIoU has increased by 4.2%. After adding WF, mIoU has increased by 1%. After adding SFHead, the overall effect increased by 0.3%.

D. *Experiment results*

The experimental results on the Vaihingen and Potsdam datasets are shown in **TABLE 2** and **TABLE 3**. Our model achieves state-of-the-art results. On the Vaihingen dataset, our proposed CUformer achieves the best F1 and mIoU. Compared with other ResNet-based backbone models, our model performs better on each class. Especially in the car category, we are 1.9 % higher than other models. Compared with the model with Transformer as the backbone, our model surpasses them in F1 and mIoU. Although we are lower than them in the segmentation accuracy of some classes, our model is significantly better than theirs when considering the segmentation efficiency and reasoning time.

As is shown in **Fig. 5**, We present a visualization of the partial segmentation results of the experiment to verify the effectiveness of our proposed model. Our model can well segment the boundary of the small object.

On the Potsdam dataset, our CUformer completely surpasses the ResNet-based backbone model. Moreover, the segmentation performance of CUformer on each class also exceeds that of Segmenter and SwinUperNet with Transformer as the backbone.

IV. CONCLUSION

In this paper, we propose Channel-wise Unet-based Transformer (CUformer) for real-time semantic segmentation of high-resolution remote sensing images. Its encoder is based on

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ResNet18 to capture multi-scale feature maps; the decoder is based on Transformer to obtain global semantic information. In order to meet the requirements of real-time segmentation tasks, we propose a channel window self-attention (CWSA) module, which reduces the computational complexity while obtaining information. In order to make full use of the efficient performance of ResNet18, we designed a weight-fused (WF) module to fuse the feature maps of ResnNet18 and CWSA. Finally, in order to refine the final prediction results, I also designed a simple fine-grained head (SFHead). In order to prove the performance of CUformer, we do a series of experiments on the Vaihingen and Potsdam datasets. The experimental results show that CUformer also achieves state-of-the-art results in high efficiency.

REFERENCES

[1] Li, R., Zheng, S., Duan, C., Wang, L., & Zhang, C. (2022). Land cover classification from remote sensing images based on multi-scale fully convolutional network. Geo-spatial information science, 25(2), 278-294.

[2] Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., ... & Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In Proceedings of the IEEE/CVF international conference on computer vision (pp. 10012-10022).

[3] Wang, L., Li, R., Zhang, C., Fang, S., Duan, C., Meng, X., & Atkinson, P. M. (2022). UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery. ISPRS Journal of Photogrammetry and Remote Sensing, 190, 196-214.

[4] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18 (pp. 234-241). Springer International Publishing.

[5] Yu, C., Gao, C., Wang, J., Yu, G., Shen, C., & Sang, N. (2021). Bisenet v2: Bilateral network with guided aggregation for real-time semantic segmentation. International Journal of Computer Vision, 129, 3051-3068.

[6] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... & Houlsby, N. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929.

[7] Oršić, M., & Šegvić, S. (2021). Efficient semantic segmentation with pyramidal fusion. Pattern Recognition, 110, 107611.

[8] Li, R., Zheng, S., Zhang, C., Duan, C., Wang, L., & Atkinson, P. M. (2021). ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of Fine-Resolution remotely sensed imagery. ISPRS Journal of Photogrammetry and Remote Sensing, 181, 84-98.

[9] Srinivas, A., Lin, T. Y., Parmar, N., Shlens, J., Abbeel, P., & Vaswani, A. (2021). Bottleneck transformers for visual recognition. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 16519-16529).

[10] Wang, L., Li, R., Wang, D., Duan, C., Wang, T., & Meng, X. (2021). Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images. Remote Sensing, 13(16), 3065.

[11] Yang, G., Zhang, Q., & Zhang, G. (2020). EANet: Edge-aware network for the extraction of buildings from aerial images. Remote Sensing, 12(13), 2161.

[12] Zhuang, J., Yang, J., Gu, L., & Dvornek, N. (2019). Shelfnet for fast semantic segmentation. In Proceedings of the IEEE/CVF international conference on computer vision workshops (pp. 0-0).

[13] Zhuang, J., Yang, J., Gu, L., & Dvornek, N. (2019). Shelfnet for fast semantic segmentation. In Proceedings of the IEEE/CVF international conference on computer vision workshops (pp. 0-0).

[14] Strudel, R., Garcia, R., Laptev, I., & Schmid, C. (2021). Segmenter: Transformer for semantic segmentation. In Proceedings of the IEEE/CVF international conference on computer vision (pp. 7262-7272).

[15] Sun, L., Cheng, S., Zheng, Y., Wu, Z., & Zhang, J. (2022). SPANet: Successive pooling attention network for semantic segmentation of remote sensing images. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 15, 4045-4057.

[16] Wang, L., Li, R., Duan, C., Zhang, C., Meng, X., & Fang, S. (2022). A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images. IEEE Geoscience and Remote Sensing Letters, 19, 1-5.

[17] Huang, Z., Liu, X., Wang, R., Zhang, M., Zeng, X., Liu, J., ... & Hu, Z. (2021). FaNet: fast assessment network for the novel coronavirus (COVID-19) pneumonia based on 3D CT imaging and clinical symptoms. Applied Intelligence, 51, 2838-2849.

[18] Ding, L., Lin, D., Lin, S., Zhang, J., Cui, X., Wang, Y., ... & Bruzzone, L. Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images. arXiv 2021. arXiv preprint arXiv:2106.15754.

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